



**ZEAL EDUCATION SOCIETY'S
ZEAL COLLEGE OF ENGINEERING AND RESEARCH
NARHE | PUNE -41 | INDIA
COMPUTER ENGINEERING DEPARTMENT**



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First Year Engineering Internship Projects Presentation
AI Resume Analyzer
Presented By:

Sr.No	Name	Roll No.
1	Darshan Matre	CO1630
2	Saayali Shinde	CO1649
3	Snehal Jagtap	CO1619
4	Shruti Khadse	CO1653
5	Snehal Raut	CO1643
6	Sanchita Patil	CO1645

Guided By: Prof. A. S. Khaple

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INTRODUCTION

- The **AI Resume Analyzer** is a web-based application that helps job seekers improve their resumes using Artificial Intelligence.
- It analyzes uploaded resumes, identifies strengths and weaknesses, checks ATS (Applicant Tracking System) compatibility, compares resumes with job descriptions, and provides personalized suggestions for improvement.
- The system aims to increase the chances of candidates getting shortlisted by recruiters while reducing the manual effort required for resume evaluation.
- The AI Resume Analyzer is developed using modern web technologies such as **HTML, CSS, JavaScript, PHP, and MySQL**, with **XAMPP** used as the local server environment.

OBJECTIVE

The main objectives of the AI Resume Analyzer are:

- ✓ To analyze resumes automatically using AI techniques.
- ✓ To evaluate resumes for ATS compatibility.
- ✓ To compare resumes with job descriptions and calculate a match score.
- ✓ To identify missing skills, keywords, and formatting issues.
- ✓ To provide improvement suggestions for better job opportunities.
- ✓ To simplify the resume screening process for both job seekers and recruiters.
- ✓ To improve the quality and professionalism of resumes.
- ✓ To reduce manual effort involved in resume screening.
- ✓ To assist HR professionals in selecting suitable candidates quickly.
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- ✓ To increase the chances of candidates getting shortlisted for interviews.

PROBLEM STATEMENT

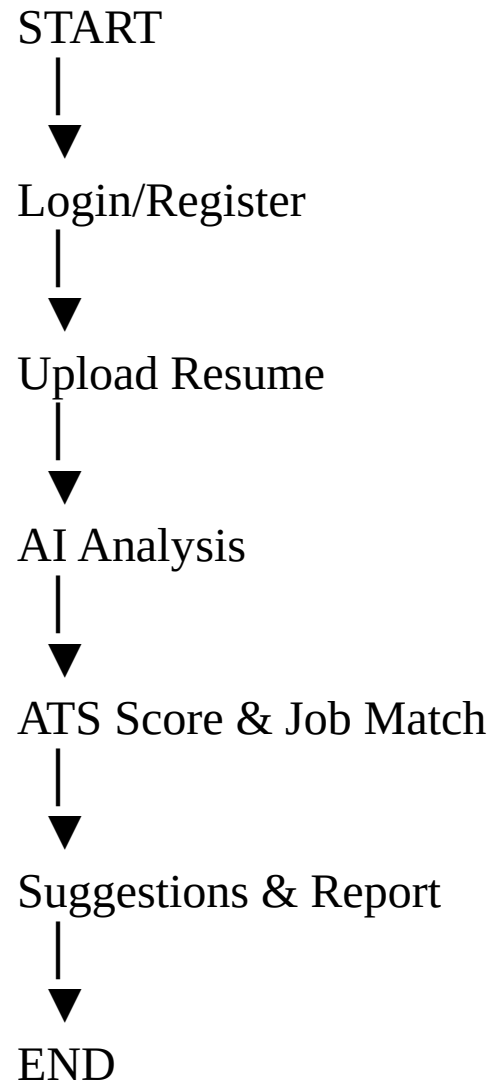
- ✓ The recruitment industry faces several challenges in evaluating resumes efficiently. Recruiters receive hundreds or even thousands of resumes for a single job opening. Manual resume screening requires significant time and effort and often results in inconsistencies and human errors.
- ✓ Many candidates possess the required skills but fail to get shortlisted because their resumes do not contain appropriate keywords or ATS-compatible formatting. Existing resume analyzers often provide limited functionality, charge expensive subscription fees, or offer generic suggestions without detailed analysis.
- ✓ Therefore, there is a need for an intelligent system that automatically analyzes resumes, compares them with job descriptions, provides ATS scores, identifies missing skills, and offers personalized recommendations. The proposed AI Resume Analyzer addresses these issues by providing an affordable, accurate, and efficient solution for resume evaluation.

MOTIVATION

The motivation behind developing this project includes:

- ✓ Increasing competition in today's job market.
- ✓ Growing use of ATS software by companies.
- ✓ High rejection rate of resumes before recruiter review.
- ✓ Need for instant resume feedback.
- ✓ Reducing dependence on professional resume writing services.
- ✓ Helping students prepare placement-ready resumes.
- ✓ Assisting fresh graduates in improving employability.
- ✓ Simplifying resume screening for HR departments.
- ✓ Using Artificial Intelligence to automate repetitive recruitment tasks.
- ✓ Improving recruitment efficiency and reducing hiring costs.
- ✓ Making resume analysis accessible through an easy-to-use web application.

FLOWCHART



SYSTEM DESIGN

System Modules

- ✓ User Authentication
- ✓ Resume Upload
- ✓ Resume Parser
- ✓ AI Analysis Engine
- ✓ ATS Score Calculator
- ✓ Skill Extraction Module
- ✓ Job Description Analyzer
- ✓ Report Generator
- ✓ Dashboard
- ✓ Contact Module
- ✓ Admin Panel

HARDWARE AND SOFTWARE REQUIREMENTS:

Hardware Requirements

- ✓ Processor: Intel Core i3 or higher
- ✓ RAM: 4 GB (8 GB Recommended)
- ✓ Storage: 20 GB Free Space
- ✓ Internet Connection
- ✓ Keyboard and Mouse

Software Requirements

- ✓ Operating System: Windows 10/11
- ✓ Frontend: HTML5, CSS3, JavaScript
- ✓ Backend: PHP
- ✓ Database: MySQL
- ✓ Server: XAMPP
- ✓ IDE: Visual Studio Code
- ✓ Browser: Google Chrome / Microsoft Edge
- ✓ Version Control: Git & GitHub

ALGORITHMS/METHODOLOGY:

Algorithms Used

- ✓ Natural Language Processing (NLP)
- ✓ Keyword Matching Algorithm
- ✓ text Parsing Algorithm
- ✓ Rule-Based ATS Evaluation
- ✓ Skill Extraction Algorithm
- ✓ Resume Scoring Algorithm

Methodology

- ✓ User registers and logs into the system.
- ✓ User uploads a resume in PDF or DOCX format.
- ✓ The system extracts text from the resume.
- ✓ AI processes the extracted content.
- ✓ Skills, education, projects, and experience are identified.
- ✓ Resume keywords are compared with job descriptions.
- ✓ ATS score is calculated.
- ✓ Suggestions for improvement are generated.

COMPARISON WITH EXISTING SYSTEMS :

Existing Resume Analyzers

- ✓ Limited ATS analysis
- ✓ Paid premium features
- ✓ Limited feedback
- ✓ Basic keyword matching
- ✓ Generic recommendations
- ✓ Limited dashboard
- ✓ Less customization

Proposed AI Resume Analyzer

Complete ATS score evaluation
Affordable and customizable
Detailed AI-based suggestions
Advanced resume-job matching
Personalized improvement tips
Interactive dashboard with reports
Easy customization for different job roles

APPLICATIONS:

The AI Resume Analyzer can be applied in:

- ✓ Colleges and Universities
- ✓ Campus Placement Cells
- ✓ Corporate HR Departments
- ✓ Recruitment Agencies
- ✓ IT Companies
- ✓ Government Recruitment Organizations
- ✓ Career Guidance Center
- ✓ Resume Writing Services
- ✓ Freelance Career Consultants
- ✓ Online Job Portals
- ✓ Professional Networking Platforms
- ✓ Skill Development Institutes
- ✓ Employee Recruitment Systems
- ✓ Internship Selection Programs
- ✓ Educational Institutions

IMPLEMENTATION SNAPSHOTS:

Snapshot	Description
Home Page	Displays the landing page with navigation menu, project introduction, features, and Login/Register options.
User Registration Page	Allows new users to create an account by entering their personal details.
User Login Page	Authenticates registered users before accessing the system.
Dashboard	Shows user profile, uploaded resumes, recent analyses, and quick navigation options.
Resume Upload Page	Enables users to upload resumes in PDF or DOCX format for analysis.
Job Description Page	Allows users to enter or upload a job description for comparison with the resume
Resume Analysis Result	Displays ATS score, extracted skills, keyword analysis, missing skills, and improvement suggestions.
Analysis Report	Shows a detailed report including resume score, job match percentage, strengths, weaknesses, and recommendations.
Contact/About Page	Provides project information and a contact form for user feedback or support.

CONCLUSION:

- ✓ The AI Resume Analyzer is an innovative web application that combines Artificial Intelligence, Natural Language Processing, and web technologies to automate resume analysis and optimization. It helps job seekers improve resume quality, achieve better ATS compatibility, and align their resumes with specific job requirements. By providing intelligent feedback, keyword analysis, and personalized recommendations, the system significantly increases the chances of candidates being shortlisted for interviews.
- ✓ The application also reduces the workload of HR professionals by automating resume screening, making the recruitment process faster, more consistent, and more efficient. With its scalable architecture, secure database, and user-friendly interface, the AI Resume Analyzer can be extended in the future to include advanced AI features such as interview preparation, career guidance, multilingual resume analysis, and AI-generated resume creation.

REFERENCES:

- ✓ HTML5 Documentation
- ✓ CSS3 Documentation
- ✓ JavaScript Documentation
- ✓ PHP Official Documentation
- ✓ MySQL Official Documentation
- ✓ XAMPP Documentation
- ✓ Git Official Documentation
- ✓ GitHub Documentation
- ✓ Applicant Tracking System (ATS) Guidelines
- ✓ Natural Language Processing (NLP) Research Papers
- ✓ Artificial Intelligence in Recruitment Research Papers
- ✓ Resume Parsing Techniques Research Papers



Thank You.....!!!!